

We are given

$$H(z) = \frac{3 - z^{-1}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}$$

I will now show every step with short, exact explanations, nothing extra.

Step 1: Find the Zeros

Definition:

Zeros are the values of z that make the **numerator** = 0.

Step 1.1: Set numerator to zero

$$3 - z^{-1} = 0$$

Step 1.2: Solve for z

$$z^{-1} = 3$$

$$z = \frac{1}{3}$$

Result

$$\text{Zero at } z = \frac{1}{3}$$

Step 2: Find the Poles

Definition:

Poles are the values of z that make the **denominator** = 0.

Step 2.1: Set denominator to zero

$$1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2} = 0$$

Step 2.2: Eliminate negative powers

Multiply the entire equation by z^2 :

$$z^2 - \frac{3}{4}z + \frac{1}{8} = 0$$

Step 2.3: Factor the quadratic

$$(z - \frac{1}{2})(z - \frac{1}{4}) = 0$$

Step 2.4: Solve

$$z = \frac{1}{2}, \frac{1}{4}$$

Result

Poles at $z = \frac{1}{2}, \frac{1}{4}$

Step 3: Inverse Z-Transform

Step 3.1: Factor denominator

$$H(z) = \frac{3 - z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})}$$

Step 3.2: Partial fraction expansion

Assume:

$$\frac{3 - z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})} = \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - \frac{1}{4}z^{-1}}$$

Step 3.3: Solve for constants

$$3 - z^{-1} = A(1 - \frac{1}{4}z^{-1}) + B(1 - \frac{1}{2}z^{-1})$$

Comparing coefficients:

$$A = 4, \quad B = -1$$

Step 3.4: Write final decomposition

$$H(z) = \frac{4}{1 - \frac{1}{2}z^{-1}} - \frac{1}{1 - \frac{1}{4}z^{-1}}$$

Step 3.5: Use standard Z-transform pair

$$Z^{-1} \left\{ \frac{1}{1 - az^{-1}} \right\} = a^n u[n]$$

Inverse Z-transform

$$h[n] = 4 \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{4}\right)^n u[n]$$

Step 4: System Properties

- **Linear:** Yes (rational $H(z)$)
 - **Time-invariant:** Yes
 - **Causal:** Yes (right-sided sequence)
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Step 5: Region of Convergence (ROC)

For a causal system, ROC is outside the outermost pole.

Outermost pole:

$$|z| = \frac{1}{2}$$

ROC

$$|z| > \frac{1}{2}$$

Step 6: Stability

Condition:

A discrete-time LTI system is stable if the ROC includes the unit circle $|z| = 1$.

Check:

$$1 > \frac{1}{2}$$

Result

Final Summary

- Zero: $z = \frac{1}{3}$
- Poles: $z = \frac{1}{2}, \frac{1}{4}$
- Impulse response:

$$h[n] = 4 \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{4}\right)^n u[n]$$

- ROC: $|z| > \frac{1}{2}$
- Causal: Yes
- Stable: Yes

now make 1 example and also solve where the system will be unstable it should be similar to the above question

Below is **one example**, very similar in form to your previous question, and it is solved **step by step**.

This time the system is **unstable**.

Example (Unstable System)

$$H(z) = \frac{2 - z^{-1}}{1 - \frac{5}{4}z^{-1} + \frac{3}{8}z^{-2}}$$

Step 1: Find the Zeros

Step 1.1: Set numerator to zero

$$2 - z^{-1} = 0$$

Step 1.2: Solve

$$z^{-1} = 2 \Rightarrow z = \frac{1}{2}$$

Zero

$$z = \frac{1}{2}$$

Step 2: Find the Poles

Step 2.1: Set denominator to zero

$$1 - \frac{5}{4}z^{-1} + \frac{3}{8}z^{-2} = 0$$

Step 2.2: Multiply by z^2

$$z^2 - \frac{5}{4}z + \frac{3}{8} = 0$$

Step 2.3: Factor

$$(z - \frac{3}{4})(z - \frac{1}{2}) = 0$$

Poles

$$z = \frac{3}{4}, \frac{1}{2}$$

Step 3: Inverse Z-Transform

Step 3.1: Factor denominator

$$H(z) = \frac{2 - z^{-1}}{(1 - \frac{3}{4}z^{-1})(1 - \frac{1}{2}z^{-1})}$$

Step 3.2: Partial fractions

$$H(z) = \frac{A}{1 - \frac{3}{4}z^{-1}} + \frac{B}{1 - \frac{1}{2}z^{-1}}$$

Step 3.3: Solve coefficients

$$2 - z^{-1} = A(1 - \frac{1}{2}z^{-1}) + B(1 - \frac{3}{4}z^{-1})$$

Matching coefficients:

$$A = 4, \quad B = -2$$

Step 3.4: Final form

$$H(z) = \frac{4}{1 - \frac{3}{4}z^{-1}} - \frac{2}{1 - \frac{1}{2}z^{-1}}$$

Step 3.5: Inverse Z-transform

$$h[n] = 4 \left(\frac{3}{4}\right)^n u[n] - 2 \left(\frac{1}{2}\right)^n u[n]$$

Step 4: System Properties

- Linear: Yes
 - Time-invariant: Yes
 - Causal: Yes
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Step 5: Region of Convergence (ROC)

For a causal system, ROC is outside the largest pole magnitude.

Largest pole:

$$|z| = \frac{3}{4}$$

ROC

$$|z| > \frac{3}{4}$$

Step 6: Stability

Stability condition:

ROC must include the unit circle $|z| = 1$.

Here:

$$1 \notin |z| > \frac{3}{4} \Rightarrow |z| = 1 \text{ is outside ROC}$$

Conclusion

System is unstable

Key Reason for Instability

The system has a **pole close to the unit circle**, and the ROC **does not include** $|z| = 1$.